

Auctions, Iterated Dominance, & Nash Equilibrium

ECON 5001
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Where We Left Off

We can now:

- separate a **frame** from a **game**
- write preferences as a complete, transitive $\succsim_i$
- represent $\succsim_i$ with an ordinal utility function
- compare two strategies of one player: **dominance**

And we found: if you have a dominant strategy, play it.

Today: what if you don't?

Log In — We're Bidding

- MobLab, today's session
- You'll be told what the item is worth **to you**
 - that number is yours alone; others have different ones
- Submit one sealed bid. Highest bid wins.
- We'll play it **twice**, under two different rules

Your payoff is your value minus what you pay, or zero if you don't win.

No talking. Bid however you think is smart.

What Did You Just Play?

Round 1 — first-price: the winner pays *her own bid*.

Round 2 — second-price: the winner pays *the highest losing bid*.

Show of hands — in which round did you bid closer to your true value?

Second-Price Auctions

The Rules

A **second-price** (or **Vickrey**) auction:

- sealed bids — nobody sees anyone else's bid
- highest bidder wins
- the winner pays the **second-highest** bid, not her own
- ties are broken by a rule fixed in advance

Why would a seller ever do this? It looks like giving money away.

An Example: The Game-Day Permit

One food truck permit for the lot outside the stadium is up for auction.

- Truck 1 has run the numbers: the permit is worth **\$30k** to her
- Bids must be one of \$10k, \$20k, \$30k, \$40k, \$50k
- Ties go to Truck 2 (decided by a coin flip beforehand)

Truck 1's payoff:

$$\pi_1 = \begin{cases} 30 - (\text{Truck 2's bid}) & \text{if Truck 1 wins} \\ 0 & \text{otherwise} \end{cases}$$

Note she pays *Truck 2's* bid, not her own.

Only Truck 1's Payoffs Shown

		Truck 2				
		10	20	30	40	50
Truck 1 (value 30)	10	0, .	0, .	0, .	0, .	0, .
	20	20, .	0, .	0, .	0, .	0, .
	30	20, .	10, .	0, .	0, .	0, .
	40	20, .	10, .	0, .	0, .	0, .
	50	20, .	10, .	0, .	-10, .	0, .

Check one cell: Truck 1 bids 30, Truck 2 bids 10.

Reading the Table

- Bidding **30** (her true value) is weakly dominant
- Bidding **40** is equivalent to bidding 30 — also weakly dominant
- Bidding **50** is strictly worse than 30. Where, and why?
- Bidding **20** or **10** is weakly worse. Where, and why?

Vickrey's Theorem

Theorem (Vickrey, 1961). In a second-price auction, if player i is selfish and greedy, then bidding her true value — $b_i = v_i$ — is a **weakly dominant strategy**.

Remarkable, because:

- it holds no matter how many bidders there are
- it holds no matter what anyone else bids
- you don't need to guess, predict, or model your rivals at all

Truth-telling isn't nice here. It's just optimal.

Why It Works: Two Cases

Let v_1 be my value and b_2 the other bid. Compare bidding v_1 to anything else.

Case 1: $b_2 \leq v_1$. Bidding v_1 wins and pays b_2 , for a payoff of $v_1 - b_2 \geq 0$.

- Bidding higher: still win, still pay b_2 . _____
- Bidding below b_2 : lose, payoff 0. _____

Case 2: $b_2 > v_1$. Bidding v_1 loses, for a payoff of 0.

- Bidding anything below b_2 : still lose. _____
- Bidding at or above b_2 : win, pay b_2 . _____

Raising your bid can only ever change *whether* you win, never *what* you pay.

The Fine Print

The theorem assumes the bidder is **selfish and greedy**.

Drop that and it fails. Suppose I also care what *you* pay:

- **Spiteful** — if I lose, I'd rather you paid more
- **Benevolent** — if I lose, I'd rather you paid less

In either case, bidding my value is no longer dominant.

Same mechanism, different preferences, different game. Again.

Back to Your Bids

Under second-price, truthful bidding is weakly dominant.

Under first-price, it is not — winning means paying your own bid, so bidding your value guarantees a payoff of exactly zero. You must **shade** down.

Round 1 (first-price), average bid/value: _____

Round 2 (second-price), average bid/value: _____

The Pivotal Mechanism

A Real Headline

Davis Enterprise, January 12, 2001:

“By consensus, the Davis City Council agreed Wednesday to order a community-wide public opinion poll to gauge how much Davis residents would be willing to pay for a park tax and a public safety tax.”

Will people answer honestly?

- If my answer doesn't affect my tax bill:
- If my answer does affect my tax bill:

The Setup

A public project (build the park) costs $\$C$ in total. There are n residents.

- Resident i 's share of the cost is c_i , with $c_1 + \dots + c_n = C$
(shares are announced in advance and need not be equal)
- Resident i starts with wealth m_i
- If built, resident i gets benefit worth v_i — possibly **negative**

$$U_i(\$m) = \begin{cases} m & \text{if the project is not built} \\ m + v_i & \text{if it is} \end{cases}$$

Efficiency says: build it if and only if $\sum_{i=1}^n v_i > C$.

Why Efficiency Isn't Enough

$n = 2$, $m_1 = 50$, $m_2 = 60$, $v_1 = 19$, $v_2 = -15$, $C = 6$, $c_1 = 6$, $c_2 = 0$.

$\sum v_i = 19 - 15 = 4 < 6 = C$, so it should *not* be built.

	Not built	Built
Resident 1	50	$50 + 19 - 6 = 63$
Resident 2	60	$60 - 15 = 45$

Gain of 13 to one, loss of 15 to the other.

The government would happily follow the rule — if it knew v_1 and v_2 .

It doesn't. And asking is exactly the problem.

Clarke's Idea

Each resident i reports a number w_i — claimed benefit (or harm, if negative).

She may lie: nothing forces $w_i = v_i$.

Decision rule:

$$\text{Build?} \quad \begin{cases} \text{Yes} & \text{if } \sum_{j=1}^n w_j > C \\ \text{No} & \text{if } \sum_{j=1}^n w_j \leq C \end{cases}$$

Then the twist. Resident i is **pivotal** if the decision *without her report* would differ from the decision *with it*.

- Not pivotal $\Rightarrow$ she pays no tax
- Pivotal $\Rightarrow$ she pays $\left| \sum_{j \neq i} w_j - \sum_{j \neq i} c_j \right|$

Worked Example

$n = 3$, $C = 10$, $c_1 = 3$, $c_2 = 2$, $c_3 = 5$. Reports: $w_1 = -1$, $w_2 = 8$, $w_3 = 3$.

$\sum w_j = 10$, which is *not* greater than $C = 10$, so the project is **not built**.

i	$\sum_{j \neq i} w_j$	$\sum_{j \neq i} c_j$	Decision without i	Tax
1	$8 + 3 = 11$	$2 + 5 = 7$	Build (pivotal)	$ 11 - 7 = 4$
2	$-1 + 3 = 2$	$3 + 5 = 8$	Don't (not pivotal)	0
3	$-1 + 8 = 7$	$3 + 2 = 5$	Build (pivotal)	$ 7 - 5 = 2$

Note residents 1 and 3 each flipped the outcome, and each pays for it.

“Then I’ll Just Avoid Being Pivotal”

You can guarantee it: report $w_i = c_i$ and you are never pivotal, never taxed.

Is that smart? Take $n = 4$, $C = 15$, $c = (5, 0, 5, 5)$. You are resident 1, with $m_1 = 40$ and $v_1 = 25$. You expect $w_2 = -40$, $w_3 = 15$, $w_4 = 20$.

Report $w_1 = c_1 = 5$: $\sum w_j = 0 < 15$, not built, and you pay nothing.

$$U_1 = 40$$

Report truthfully, $w_1 = 25$: $\sum w_j = 20 > 15$, so it *is* built. You are pivotal, taxed $|(-5) - (10)| = 15$.

$$U_1 = 40 + 25 - 5 - 15 = 45$$

Clarke's Theorem

Theorem (Clarke, 1971). In the pivotal mechanism, reporting truthfully — $w_i = v_i$ — is a **weakly dominant strategy** for every player.

Same shape as Vickrey: the tax you pay is built out of *other people's* numbers, so your report moves the decision without moving your own bill.

Log In Again — Pick a Number

Choose an integer from **1** to **100**.

Whoever is closest to $\frac{2}{3} \times (\text{the average of everyone's number})$ wins.

No talking. We'll play a few rounds.

Results

	Average	Winning number
Round 1:	_____	_____
Round 2:	_____	_____
Round 3:	_____	_____

What happened between rounds? And why?

Iterated Deletion

The Problem

If a player has a dominant strategy, she should play it.

But most games don't have one. So what then?

New idea: even without a dominant strategy, a rational player will never play a **strictly dominated** one.

And if I know you're rational, I know you won't either — which shrinks the game I'm actually facing.

And then *that* smaller game may have dominated strategies of its own.

Iterated Deletion of Strictly Dominated Strategies.

Start with game G .

- G^1 : delete every strictly dominated strategy, for every player
- G^2 : do it again, inside G^1
- $\vdots$
- G^∞ : the output, reached in finitely many steps if G is finite

If G^∞ is a single strategy profile, we call it the **iterated strict dominant-strategy profile**.

Otherwise it's just "the output of IDSDS."

Order doesn't matter: any sequence of deletions of strictly dominated strategies reaches the same output.

Doing It

		Player 2		
		L	C	R
Player 1	T	4, 3	5, 1	6, 2
	M	2, 1	8, 4	3, 6
	B	3, 0	9, 6	2, 8

Nobody has a dominant strategy here. Check a couple:

- T vs. M for Player 1: $4 > 2$, but $5 < 8 \Rightarrow$ neither dominates
- T vs. B for Player 1: $6 > 2$, but $5 < 9 \Rightarrow$ neither dominates

So start with Player 2.

Doing It

Step 1. For Player 2, R strictly dominates C : $2 > 1$, $6 > 4$, $8 > 6$.
Delete C .

		Player 2	
		L	R
Player 1	T	4, 3	6, 2
	M	2, 1	3, 6
	B	3, 0	2, 8

Step 2. Now T strictly dominates *both* M and B for Player 1 — which it did not do a moment ago. Delete them.

		Player 2	
		L	R
Player 1	T	4, 3	6, 2
	B	5, 4	7, 3

Step 3. Player 2 now faces a one-row game: L gives 3, R gives 2. Delete R .

$$G^{\infty} = \{(T, L)\}$$

Nobody had a dominant strategy at the start, and yet the game has a unique prediction — *provided* each player knows the other is rational, and knows that the other knows, and so on.

Bonanno's Example

Four rounds instead of three. Only the deleted strategy is named each step.

		Player 2			
		e	f	g	h
Player 1	A	6, 3	4, 4	4, 1	3, 0
	B	5, 4	6, 3	0, 2	5, 1
	C	5, 0	3, 2	6, 1	4, 0
	D	2, 0	2, 3	3, 3	6, 1

1. delete h (dominated by g)
2. delete D (dominated by C)
3. delete g (dominated by f)
4. delete C (dominated by A)

Output: $\{(A, e), (A, f), (B, e), (B, f)\}$ not a single profile

Back to Your Numbers

$\frac{2}{3}$ of the average can never exceed $\frac{2}{3} \times 100 \approx 66.7$.

- So 67 strictly dominates everything above it. Delete 68–100.
- Now the target can't exceed $\frac{2}{3} \times 67 \approx 44.7$. Delete 45–67.
- Now it can't exceed $\frac{2}{3} \times 45 = 30$. Delete 31–45.
- Keep going ... where does it end?

So why didn't we all play that in round 1?

IDWDS, and Why It's Trickier

What if we delete **weakly** dominated strategies too?

		Player 2	
		L	R
Player 1	A	4, 0	0, 0
	T	3, 2	2, 2
	M	1, 1	0, 0
	B	0, 0	1, 1

Two perfectly reasonable places to start:

- M is strictly dominated by T ($3 > 1, 2 > 0$)
- B is strictly dominated by T ($3 > 0, 2 > 1$)

Order One: Delete M First

Delete $M \Rightarrow$ for Player 2, R now weakly dominates $L \Rightarrow$ delete L

		Player 2	
		R	
Player 1	A	0, 0	
	T	2, 2	
	B	1, 1	

Now A and B are strictly dominated by T .

Answer: (T, R) , payoffs $(2, 2)$

Order Two: Delete B First

Delete $B \Rightarrow$ for Player 2, L now weakly dominates $R \Rightarrow$ delete R

		Player 2	
		L	R
Player 1	A	4, 0	0, 4
	T	3, 2	2, 3
	M	1, 1	1, 1

Now T and M are strictly dominated by A .

Answer: (A, L) , payoffs $(4, 0)$

So the Procedure Isn't Well Defined

Two legitimate orders, two different answers — and not by a little: $(2, 2)$ versus $(4, 0)$.

Fix (Definition). At every step, identify *all* weakly or strictly dominated strategies for *all* players, and delete them **simultaneously**.

Applied to that game, IDWDS gives $\{(A, L), (A, R), (T, L), (T, R)\}$.

Note this is not a single profile — so the game has no iterated weak dominant-strategy solution.

A Word of Caution

IDSDS has a clean story: rationality, plus knowing others are rational, and so on.

IDWDS does not.

- To delete a *weakly* dominated strategy, you must believe your opponent might play anything — otherwise the tie-breaking case never arises
- But deletion is precisely the act of ruling strategies out

So the justification needs rationality *and* a notion of caution, and the two pull against each other.

We won't resolve that here. Just don't treat the two procedures as equally innocent.

Nash Equilibrium

Still Not Enough

Games solved outright by IDSDS or IDWDS are the exception.

We need something that applies to *every* game.

The idea: stop asking “what will a rational player do?” and start asking

“which profiles are **stable**?”

Definition: Nash Equilibrium (Two Players)

A strategy profile $s^* = (s_1^*, s_2^*)$ is a **Nash equilibrium** if

1. for every $s_1 \in S_1$: $\pi_1(s_1^*, s_2^*) \geq \pi_1(s_1, s_2^*)$
2. for every $s_2 \in S_2$: $\pi_2(s_1^*, s_2^*) \geq \pi_2(s_1^*, s_2)$

In words: **holding the other player's strategy fixed**, neither player can do better by changing her own.

Note what's fixed and what varies in each line. That's the whole definition.

Four Ways to Say It

No regret. After seeing what the other player did, nobody wishes she had chosen differently.

Self-enforcing agreement. If the players could talk beforehand and shake on s^* , nobody would want to break it — assuming the other keeps it.

Viable recommendation. If a third party publicly recommended s^* , nobody would want to deviate — assuming the others comply.

Rationality with correct beliefs. Each player best-responds to what she believes the other will do, and those beliefs are right.

These are all the same inequalities in different clothes.

Finding Them

		Player 2		
		L	C	R
Player 1	T	3, 2	0, 0	1, 1
	M	3, 0	1, 5	4, 4
	B	1, 0	2, 3	3, 0

Check (T, L) :

- Player 1, holding L fixed: T gives 3; M gives 3; B gives 1 ✓
- Player 2, holding T fixed: L gives 2; C gives 0; R gives 1 ✓

So (T, L) is a Nash equilibrium. Find the other one.

Best Replies

Strategy s_i is a **best reply** to s_{-i} if

$$\pi_i(s_i, s_{-i}) \geq \pi_i(s'_i, s_{-i}) \quad \text{for every } s'_i \in S_i$$

Equivalent definition of Nash equilibrium:

s is a Nash equilibrium $\iff$ every s_i is a best reply to s_{-i}

This is the version worth memorizing. It also tells you how to *find* equilibria.

The Underlining Trick

1. In each **column**, underline Player 1's largest payoff (all ties)
2. In each **row**, underline Player 2's largest payoff (all ties)
3. Any cell with **both** numbers underlined is a Nash equilibrium

		Player 2			
		E	F	G	H
Player 1	A	4, 0	3, 2	2, 3	4, 1
	B	4, 2	2, 1	1, 2	0, 2
	C	3, 6	5, 5	3, 1	5, 0
	D	2, 3	3, 2	1, 2	3, 3

Your turn. Columns first, then rows.

The Underlining Trick

		Player 2			
		E	F	G	H
Player 1	A	<u>4</u> , 0	3, 2	2, <u>3</u>	4, 1
	B	<u>4</u> , <u>2</u>	2, 1	1, <u>2</u>	0, <u>2</u>
	C	3, <u>6</u>	<u>5</u> , 5	<u>3</u> , 1	<u>5</u> , 0
	D	2, <u>3</u>	3, 2	1, 2	3, <u>3</u>

Only (B, E) has both payoffs underlined — the unique Nash equilibrium.

Watch the near-misses:

- (A, E): Player 1 is happy, Player 2 would rather move to G
- (C, F): Player 1 is happy, Player 2 would rather move to E

More Than Two Players

A profile $s^* \in S$ is a Nash equilibrium if, for every player i ,

$$\pi_i(s^*) \geq \pi_i(s_1^*, \dots, s_{i-1}^*, s_i, s_{i+1}^*, \dots, s_n^*) \quad \text{for all } s_i \in S_i$$

n inequalities, one per player. Nothing new — just more of them.

For three players we draw one table per strategy of Player 3.

- underline Players 1 and 2 as before, within each table
- underline Player 3's payoff if it's her largest *in that cell across tables*

Two Theorems

Let $NE(G)$, $IDSDS(G)$, $IDWDS(G)$ be the respective outputs.

Theorem. For every ordinal strategic-form game G ,

$$NE(G) \subseteq IDSDS(G)$$

- A Nash equilibrium never uses a strategy that IDSDS deletes
- So IDSDS never throws away an equilibrium — it's safe
- But the same is **not** true of IDWDS: it is possible that $NE(G) \neq \emptyset$ and yet $NE(G) \cap IDWDS(G) = \emptyset$

Theorem. If $IDSDS(G) = \{s\}$ then s is a strict Nash equilibrium.
If $IDWDS(G) = \{s\}$ then s is a Nash equilibrium.

A Bigger Example

50 players. Each secretly writes a request: a multiple of \$10, up to \$100.

- If **5 or fewer** players ask for \$100, everyone gets what they asked for
- Otherwise, everyone gets nothing

All players are selfish and greedy. What are the Nash equilibria?

A Bigger Example

There are two families of equilibria.

(a) Seven or more players request \$100.

Everyone gets nothing. Can anyone gain by switching alone?

(b) Exactly five request \$100, and all 45 others request \$90.

Everyone is paid. Can anyone gain by switching alone?

Convince yourself that *exactly six* requesting \$100 is not an equilibrium.

When There Is None

Matching Pennies. Both match $\Rightarrow$ Player 1 wins; otherwise Player 2 wins.

		Player 2	
		H	T
Player 1	H	<u>1</u> , 0	0, <u>1</u>
	T	0, <u>1</u>	<u>1</u> , 0

No cell has both payoffs underlined.

With only ordinal payoffs, a game need not have **any** Nash equilibrium.

Fixing this requires letting players randomize — and randomizing requires cardinal payoffs. That's Chapter 5.

Infinite Strategy Sets

No Table, Same Definition

Two players. Each writes down a real number ≥ 1 , so $S_1 = S_2 = [1, \infty)$.

Player 1 writes x , Player 2 writes y :

$$\pi_1(x, y) = \begin{cases} x - 1 & \text{if } x < y \\ 0 & \text{if } x \geq y \end{cases} \quad \pi_2(x, y) = \begin{cases} y - 1 & \text{if } x > y \\ 0 & \text{if } x \leq y \end{cases}$$

You want to write the larger... no, the *smaller* number. But not too small.

What are the Nash equilibria?

The Answer Is $(1, 1)$ — and Only $(1, 1)$

It is an equilibrium. At $(1, 1)$ both get 0. If Player 1 switches to any $x > 1$, then $x \geq y = 1$, so she still gets 0.

Nothing else is.

- If $x = y > 1$: Player 1 gets 0, but any $\hat{x}$ with $1 < \hat{x} < x$ gives $\hat{x} - 1 > 0$
- If $x < y$: Player 1 gets $x - 1$, but any $\hat{x}$ with $x < \hat{x} < y$ gives $\hat{x} - 1 > x - 1$
- If $y < x$: same argument for Player 2

And now the strange part. For Player 1, $x = 1$ is **weakly dominated** by every other strategy.

So the unique Nash equilibrium has both players using a weakly dominated strategy.

Cournot Competition

Two firms produce an identical good. Firm i chooses quantity $q_i \geq 0$ at constant unit cost c . Price is set by total output:

$$P = a - b(q_1 + q_2), \quad a > c, \quad b > 0$$

Firm 1's profit:

$$\pi_1(q_1, q_2) = Pq_1 - cq_1 = (a - c)q_1 - bq_1^2 - bq_1q_2$$

Strategy sets are $[0, \infty)$ — infinite. But a Nash equilibrium is still just a pair $(\bar{q}_1, \bar{q}_2)$ where neither firm wants to move.

Cournot wrote this down in 1838 — inventing Nash equilibrium about 120 years early, for this one class of games.

Solving It

Fix $\bar{q}_2$. Then π_1 is a function of q_1 alone, and it's concave ($\partial^2 \pi_1 / \partial q_1^2 = -2b < 0$), so set the derivative to zero:

$$\frac{\partial \pi_1}{\partial q_1} = a - c - 2b\bar{q}_1 - b\bar{q}_2 = 0$$

$$\frac{\partial \pi_2}{\partial q_2} = a - c - 2b\bar{q}_2 - b\bar{q}_1 = 0$$

Two equations, two unknowns:

$$\bar{q}_1 = \bar{q}_2 = \frac{a - c}{3b}$$

Cournot: The Numbers

$$\text{At } \bar{q}_1 = \bar{q}_2 = \frac{a - c}{3b}:$$

- Price: $P = \frac{a + 2c}{3}$

- Each profit: $\frac{(a - c)^2}{9b}$

Try it: $a = 25$, $b = 2$, $c = 1$.

$$\bar{q}_1 = \bar{q}_2 = \underline{\hspace{2cm}} \quad P = \underline{\hspace{2cm}} \quad \text{profit} = \underline{\hspace{2cm}}$$

Where We Landed

- **Second-price auction:** truthful bidding is weakly dominant (Vickrey)
- **Pivotal mechanism:** truthful reporting is weakly dominant (Clarke)
- **IDSDS:** delete strictly dominated strategies, repeat — order-free, justified by common knowledge of rationality
- **IDWDS:** order matters, so delete simultaneously — and the justification is murkier
- **Nash equilibrium:** nobody gains by deviating alone
- $NE \subseteq IDSDS$, but a game may have no Nash equilibrium at all

Next: what happens when players can randomize.

Reading & Practice

- Bonanno, *Game Theory*, §2.3–2.7
- Exercises: §2.9.3 (auctions), §2.9.4 (pivotal mechanism), §2.9.5 (iterated deletion), §2.9.6 (Nash equilibrium), §2.9.7 (infinite games)
- Proofs of both theorems are in §2.8
- Solutions in §2.10 — try them first